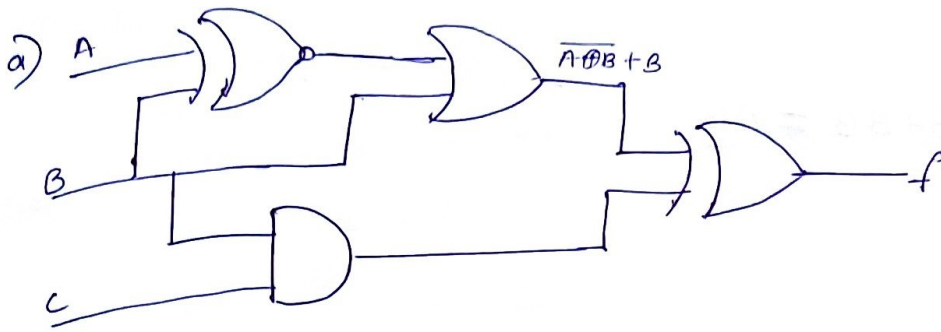


Boolean Algebra

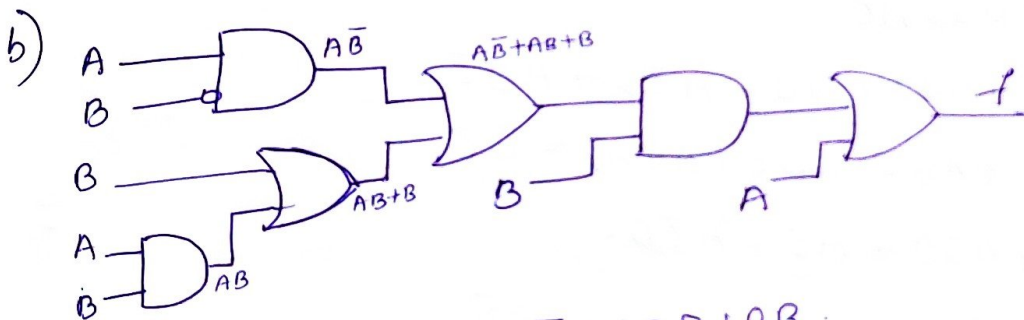
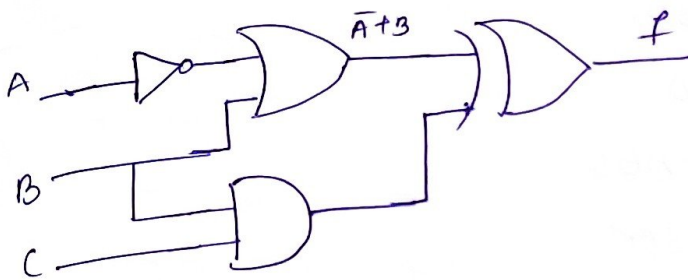
1. Derive output expression and draw simplified circuits.



$$\Rightarrow \begin{aligned} \overline{A \oplus B} + B &= \overline{A \oplus B} + B \\ &= \overline{A \oplus B} + B = (B + \overline{A})(B + \overline{B}) \} \text{OR gate} \\ &= \underline{\underline{\overline{A} + B}} \end{aligned}$$

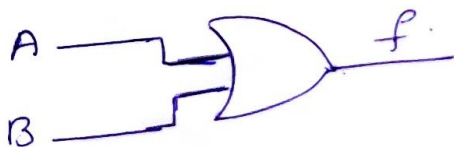
$$y = BC \text{ (AND gate)}$$

$$f = \underline{\underline{(\overline{A} + B) \oplus BC}} \text{ (XOR)}$$

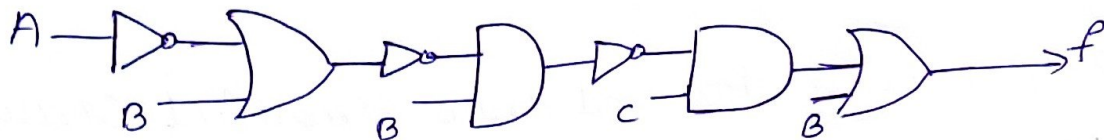


$$\begin{aligned} (\overline{A}B + AB + B)B &= \overline{A}B\overline{B} + AB\overline{B} + BB \\ &= 0 + A + B \end{aligned}$$

$$f = A + B + A = \underline{\underline{A + B}}$$



c)



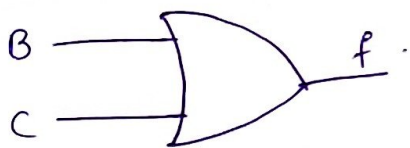
$$= \bar{A} + B$$

$$= (A + \bar{B})B = AB + B\bar{B} = \underline{AB}$$

$$= \bar{A}B \cdot C$$

$$= \bar{A}B \cdot C + B$$

$$= \underline{B + C}$$



2) Minimize the boolean expressions

a) $\bar{A}C\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

$$f = E(\bar{A}\bar{D} + AB) + \bar{A}\bar{B}\bar{D} + ABD$$

$$f = CE(\bar{A}\bar{D} + AB) + \bar{A}\bar{B}\bar{D} + ABD$$

This is in its minimum SOP

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + A\bar{B}\bar{C}$

$$\bar{A}C + \bar{A}C\bar{D} \Rightarrow \bar{A}C$$

$$F = \bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D + A\bar{B}\bar{C}$$

$$= \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} = \bar{B}\bar{C}(A + \bar{A}) = \bar{B}\bar{C}$$

$$F = \bar{B}\bar{C} + ABD + \bar{A}C + A\bar{C}D$$

c) $= \bar{A}\bar{C} + DC + \bar{A}\bar{B} + \bar{A}BC + A\bar{C}\bar{D} + B\bar{C}\bar{D}$

$$\Rightarrow BC + \bar{A}BC = BC(1 + \bar{A}) = BC$$

$$F = \bar{A}\bar{C} + \bar{A}\bar{B} + BC + A\bar{C}\bar{D} + B\bar{C}\bar{D}$$

$$d) \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_1 \bar{x}_3 x_4 + x_1 \bar{x}_3 x_4 + x_2$$

$$= x_2 + x_1 x_2 \bar{x}_3 x_4 = x_2 (1 + x_1 \bar{x}_3 x_4) = x_2$$

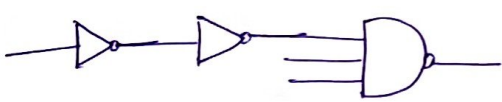
$$F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_2$$

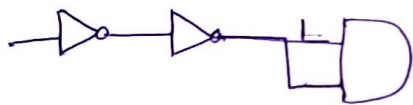
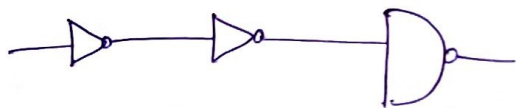
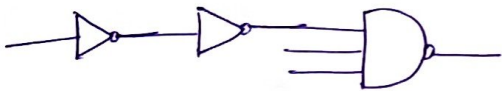
$$x_2 + \bar{x}_2 \bar{x}_1 x_4 = x_2 + \bar{x}_1 x_4$$

$$F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + x_2 + \bar{x}_1 x_4$$

$$= \bar{x}_1 (\bar{x}_3 \bar{x}_4 + x_4) + x_3 x_4 + x_3 x_4 + x_2$$

$$= \bar{x}_1 x_4 + \bar{x}_1 \bar{x}_2 + x_3 x_4 + x_2$$

3) which single gate can represent the functionality of  give circuit?



$$y_1 = \overline{A \bar{B} \bar{C}}, y_2 = \overline{A \bar{B} C}, y_3 = \overline{A \bar{B} C}, y_4 = \overline{A \bar{B} C}$$

$$y_1 = \bar{A} + B + C, y_2 = A + \bar{B} + C, y_3 = A + B + \bar{C}, y_4 = A + \bar{B} + \bar{C}$$

$$y = [(\bar{A} + B + C) \cdot (A + \bar{B} + C) \cdot (A + B + \bar{C}) \cdot (A + \bar{B} + \bar{C})]'$$

$$y = A \oplus B \oplus C$$

$\therefore$ It is a XOR gate

4) K-map

$$a) f(a, b, c, d) = \sum_m(4, 6, 8, 10, 11, 12, 15) + \sum_d(3, 5, 7, 14)$$

ab \ cd	00	01	11	10
00	0	0	X	0
01	1	X	X	1
11	1	0	1	0
10	1	X	1	1

$$f = \underline{\underline{b \bar{c} d + a \bar{b} + cd + \bar{a} b}}$$

b) $f(a, b, c, d) = \prod_m(0, 2, 4, 6, 7, 9), \prod_d(10, 11)$

ab \ cd				
	00	01	11	10
00	0			0
01	0		0	0
11				
10		0	X	X

$$f = (a+d)(a+\bar{b}+\bar{c})(\bar{a}+b+d)$$